

IEW Hackathon Problem: Application of AI & ML in the exploration of hydrocarbons, accelerating data analysis for predicting hydrocarbon presence, interpreting seismic data, and optimizing drilling locations.

Title: Multi-Agent AI System for Risk-Aware, Explainable Hydrocarbon Exploration

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Introduction

Hydrocarbon exploration has entered a new era characterized by massive, heterogeneous data influx (seismic, well logs, production, economics, ESG metrics), rising risk, regulatory requirements, and the pressure to minimize financial and environmental losses. As highlighted in recent reviews [1,2], this complexity is especially acute in unconventional reservoirs, where geological heterogeneity, incomplete data, and inter-disciplinary constraints challenge traditional empirical methods and even most existing single-model ML deployments.

Research Gaps and Novelty of the Proposed Framework

Contemporary AI deployments often face fragmented workflows, lack of physical interpretability, high model uncertainty, and a tendency to overfit or under-perform when applied outside their training zones. This proposal introduces a multi-agent, hybrid-AI system (AEIS) that addresses these gaps through:

Agent Specialization: Each agent (seismic, petrophysical, economic, risk/ESG) combines optimal state-of-the-art architectures—**CNNs with self-attention for seismic, ensemble hybrid MLP/RF/XGBoost for petrophysical, and explainable economic and risk models**—each tuned to the unique challenges of upstream exploration.

Explicit Uncertainty and Interpretability: Every agent delivers robust uncertainty quantification using **Monte Carlo dropout, SHAP, and physical-constrained hybrid modeling**, while a final consensus engine provides explainable, auditable outputs through scenario analytics and SHAP-based feature importance.

Workflow Integration: AEIS fuses data-driven predictions with geoscience priors, advanced data preprocessing, physical-constrained modeling, and human-centric, interactive analytics, directly addressing current industrial and academic critiques

ESG/Beyond-Compliance: An environmental/risk veto agent incorporates real-time regulatory rules and carbon/water footprint data, joining economic and technical outcomes.

Methodology with System Architecture Overview

A. Data Acquisition, Management, and Preprocessing

- Integrate 3D seismic cubes, well logs, economics (costs, commodity pricing), and regulatory/ESG constraints using standardized, multi-scale data formats.
- Carry out rigorous cleaning, normalization, handling of missing data, and synthetic edge-case generation.
- Employ feature engineering (extracted seismic attributes, advanced log-derived indicators) and physical constraints (e.g. fluid flow, geomechanics) to anchor models in reservoir physics.

B. Agent Architecture and Model Design

Problem/Module	Methods/Models	Features
Seismic Interpretation	CNN + Self-Attention, TL, MPSO-ANN	Probabilistic maps, uncertainty quantification, transfer learning for cross-field adaptation
Lithofacies & Petrophys.	Hybrid: CNN, XGBoost, SVM, RF, STNet	Ensemble/hybrid and hierarchical models to maximize robustness and interpretability; SHAP for feature importance analysis
Reservoir Prediction	Deep-Residual CNN, Hybrid MLP/ELM	Data fusion (well logs, seismic); GA/PSO for hyperparameter tuning
Economic/ESG/Risk	Bayesian/Monte Carlo, Veto agent	Real-time NPV, ROI, carbon/water impact, regulatory compliance; scenario-based veto logic

C. Multi-Agent Consensus and Explainability

- Weighted voting based on agent confidence (epistemic/aleatoric/structural) determines each recommendation, with hard veto from risk/environmental agent.

- Each agent provides SHAP-based and scenario-based explanations for its decision, reporting both prediction and underlying drivers.
- Visualization layer: 3D probability/risk overlays, time/space scenario dashboard, chatbot for natural language scenario simulation.

D. Validation and Industrialization

- Validate on historical and synthetic holdout blocks in real-field benchmarks.
- Metrics: drilling success rate, NPV uplift, decision latency, ESG compliance, interpretability score.
- Embedding physical constraints ensures predictions align with geoscientific expectations.

Expected Outcomes and Impact

- Higher drilling success and lower non-productive risk than single-model or non-integrated ML deployments.
- Transparency and confidence for operators (providing auditable recommendations with explicit risk quantification and rationale).
- Faster, more adaptive decision cycles (moving from months to hours/days).
- End-to-end compliance, risk mitigation, and knowledge transfer across technical and regulatory boundaries, making advanced workflows accessible across the value chain from supermajors to smaller operators.

References

- Lawal, Ahmad et al., *Machine Learning in Oil and Gas Exploration: A Review*, IEEE Access, 2024
- Chen, Feiyu et al., *A Review of AI Applications in Unconventional Oil and Gas Exploration and Development*, Energies, 2025